

Northshore University Hospital — Department of Pathology

Neuropathology / Molecular Diagnostics

Patient:	[REDACTED, M.] (MRN 87432)
Age / Sex:	47 / F
Accession:	S26-018472
Specimen received:	2026-04-08
Reported:	2026-04-08
Clinician:	Dr. K. Holst, Neurology

FINAL DIAGNOSIS

Astrocytoma, IDH-mutant, CNS WHO grade 3
Per WHO CNS5 (2021).

CLINICAL HISTORY

47-year-old woman with 6-week history of new-onset focal motor seizures. MRI brain: 3.8 cm right frontal lobe mass, patchy enhancement, moderate vasogenic edema. No prior cancer hx. No family hx of CNS tumors. Underwent gross total resection on 2026-04-04.

SPECIMEN

Right frontal lobe mass, gross total resection.

GROSS DESCRIPTION

Multiple fragments of soft gray-tan tissue, aggregate 4.1 x 3.2 x 2.6 cm, with focal hemorrhage. No grossly necrotic or cystic areas. Entirely submitted in cassettes A1-A12.

MICROSCOPIC DESCRIPTION

Sections demonstrate a moderately to markedly hypercellular astrocytic proliferation infiltrating cerebral cortex and subcortical white matter. The tumor cells exhibit moderate nuclear pleomorphism with elongated, irregular nuclei and visible nucleoli; cytoplasm is scant to moderate, eosinophilic. Mitotic figures are readily identified at up to 4 per 10 HPF in the most cellular areas.

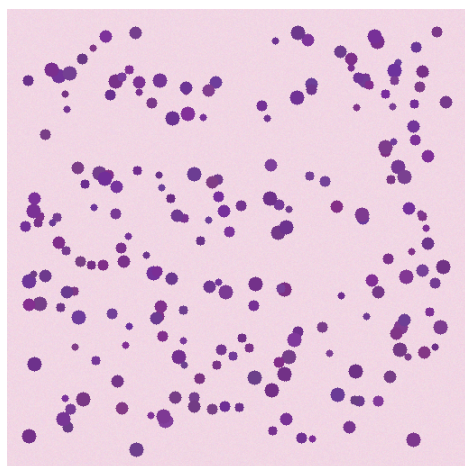
Microvascular proliferation and tumor necrosis are NOT identified. No oligodendroglial morphology is appreciated. The infiltration pattern is consistent with a diffuse glioma rather than a circumscribed entity. Reactive astrocytosis and scattered macrophages are present at the tumor-brain interface.

IMMUNOHISTOCHEMICAL STAINS

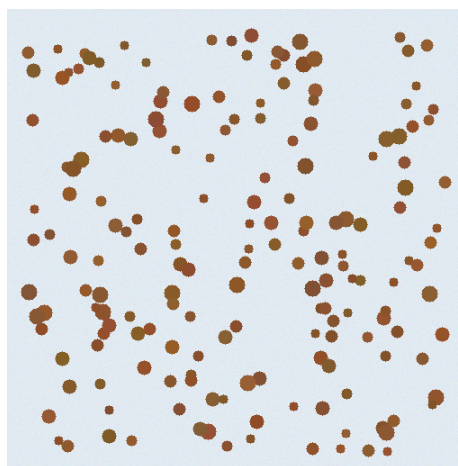
GFAP: Diffuse, strong positivity in tumor cells. **IDH1 R132H:** Strong cytoplasmic positivity. **ATRX:** Loss of nuclear expression in tumor cells (retained in vessels — internal control). **p53:** Strong nuclear positivity in >75% of tumor cells (mutant pattern).

Ki-67 (MIB-1): ~15% in hotspots; ~5% overall. **Olig2:** Positive. **EMA:** Negative. **Synaptophysin:** Negative in tumor cells.

REPRESENTATIVE PHOTOMICROGRAPHS



H&E; ×20 — frontal lobe infiltrative astrocytic tumor



IDH1 R132H IHC ×20

MOLECULAR FINDINGS

Targeted NGS panel (in-house): 523 cancer-associated genes; SNV/indel/CNV. Variants tiered per AMP/ASCO/CAP 2017. See assay methodology, appendix A.

Sequence variants (SNV/indel):

Gene	HGVSc	HGVSp	VAF	Classification
IDH1	c.395G>A	p.R132H	0.42	Pathogenic
TP53	c.524G>A	p.R175H	0.38	Pathogenic
ATRX	c.6018del	p.K2007Nfs*4	0.41	Pathogenic

Chromosomal / structural:

- 1p/19q non-codeleted (intact) — by FISH on cassette A6 (*chromosomal*)

Copy-number alterations of interest:

- CDKN2A/B** — copy number 2 (no homozygous deletion)

MSI status:	MSS (stable)
TMB (mut/Mb):	2.4
MGMT promoter (pyrosequencing, 4 CpGs averaged):	Unmethylated (mean 4.1% — cutoff <10%)

INTERPRETATION / COMMENT

Findings are diagnostic of a diffuse astrocytic glioma with IDH1 R132H mutation, ATRX loss, and TP53 alteration. The histologic features (mitotic activity present, no MVP and no necrosis) place this at the upper end of CNS WHO grade 3 per current criteria. The molecular profile (IDH-mutant, 1p/19q intact, ATRX loss) excludes oligodendroglioma. MGMT promoter is unmethylated, which has therapeutic implications. No EGFR amplification, +7/-10 signature, or TERT promoter mutation was detected; the case is NOT consistent with IDH-wildtype glioblastoma. Multidisciplinary tumor board discussion is recommended.

Electronically signed by **Dr. M. Reyes, MD, PhD**, Neuropathologist (2026-04-08)

Reviewed with: Dr. S. Ono, MD (Molecular Pathology)

Trainee: Dr. P. Iyer (PGY-3, neuropathology rotation)

ADDENDUM (2026-04-13)

MGMT promoter methylation by pyrosequencing returned 4.1% (unmethylated, cutoff <10%). Result added to molecular section. No change to integrated diagnosis.

Electronically signed by Dr. M. Reyes, MD, PhD — addendum

* This report reflects the state of laboratory science at the date of sign-out. Variants are tiered per AMP/ASCO/CAP 2017. Methylation by pyrosequencing; cutoff <10% considered unmethylated. See institutional policy LAB-SOP-NGS-002 for performance characteristics.